

IntelliAuth™ / OptumAuth-AI: Complete 14-Artifact Production Portfolio

Exhaustive End-to-End Clinical AI Product Documentation: Strategy, PRD, Architecture, Evals, Safety FMEA, Clinical Trials & SRE

Operating Standard

UHG / Optum / UMR / Epic

Compliance Authority

CMS-0057-F / HIPAA / ERISA

Net Annual P&L Value

\$1,850,000 / Operating Region

Production Status

Live in Assisted & Fast-Track GA

Complete 14-Document Enterprise Documentation Taxonomy

ENTERPRISE AI PRODUCT DOCUMENTATION MESH (14 ARTIFACTS)		
1. Product & Strategy - Doc 1: Exec ROI Memo - Doc 2: Master PRD - Doc 12: RICE Backlog	2. Architecture & Data - Doc 3: Tech Design (TDD) - Doc 4: Interface Control (ICD & FHIR Da Vinci)	3. Clinical Safety & Risk - Doc 5: Clinical FMEA (ISO 14971) - Doc 6: Statistical Analysis Plan (SAP) - Doc 8: Clinical SOP & HITL Guide
4. AI & Model Governance - Doc 7: NIST Model Card - Doc 13: Evals Spec & CI/CD Gates	5. Security & Compliance - Doc 11: CMS-0057-F Audit - HIPAA Safe Harbor Enclave & ERISA Addendum	6. Operations & Strategy - Doc 9: Phased Rollout & GTM Runbook - Doc 10: SRE Observability & Incident Play - Doc 14: JTBD & Strategic Moat Analysis

DOCUMENT 1 OF 14

Executive Business Case & Financial ROI Memo

Audience: Chief Medical Officer (Optum), SVP Health Plan Ops (UHC), VP Utilization Management (UMR)

1. Executive Summary & Problem Baseline

Prior Authorization (PA) review across UnitedHealthcare Commercial, Medicare Advantage, and UMR self-funded employer accounts represents a severe administrative bottleneck. Manual processing across 48,000 regional encounters averages **72.0 hours in turnaround time (TAT)**, consuming 18 FTE Utilization Management nurses (\$1,420,000 annual labor spend) with a **25.8% first-pass denial rate** driven by missing clinical documentation, costing \$420,000 annually in ERISA appeal rework.

2. Production Realized Outcomes & P&L Financial Model

ENTERPRISE METRIC	PRE-LAUNCH BASELINE (MANUAL)	PRODUCTION TARGET SLA	MEASURED PRODUCTION RESULT
Prior Auth Turnaround (TAT)	72.0 Hours (3–5 Days)	< 15.0 Minutes	8.2 Minutes -88.6% Total Latency
Nurse Active Touch-Time	28.5 Minutes / Case	< 5.0 Minutes / Case	3.4 Minutes / Case -88.0% Labor Compression
First-Pass Clean Approval Rate	74.2%	> 92.0%	95.8% +21.6% Approval Lift
Nurse Daily Throughput	16 Cases / Nurse / Day	60 Cases / Day	78 Cases / Nurse / Day 4.8x Expansion
Appeals & Resubmission Spend	\$420,000 / Year	< \$80,000 / Year	\$52,000 / Year -87.6% Sunk Cost Reduction
Net Annual Value Realization	—	\$1,500,000 / Year	\$1,850,000 / Year +23% vs Target ROI

Direct Labor Savings: (18 FTEs - 5 FTEs) * \$78,888 = \$1,025,544 / Year

Appeals Cost Reduction: \$420,000 - \$52,000 = \$368,000 / Year

Prevented Denial Reimbursement Value: \$825,000 / Year

AI Cloud Infrastructure Costs: (48,000 cases * \$0.11 LLM/VPC cost) + \$30,000 Vector DB/APM = -\$35,280 / Year

Net Annual Operational Value: \$1,025,544 + \$368,000 + \$825,000 - \$35,280 = **\$2,183,264 Gross / \$1,850,000 Risk-Adjusted Net**

Master Product Requirements Document (PRD): IntelliAuth Engine

Audience: Staff ML/NLP Engineers, Integration Architects, Clinical Informaticists, Product Designers

1. Functional Requirements (FR)

- **FR-1 (HL7 Da Vinci Ingestion):** Continuously ingest encounter payloads via order-sign CDS Hooks in Epic/Cerner. Pull FHIR R4 resources: Condition, Procedure, Observation, DiagnosticReport, and DocumentReference.
- **FR-2 (Dynamic Multi-Tenant Plan Engine):** Ingest and index active InterQual Criteria, CMS LCDs/NCDs, and UHC Medical Policies. Dynamically resolve UMR Employer Group exclusions by matching member Group ID against employer Summary Plan Descriptions (SPDs).
- **FR-3 (Deterministic Extraction with Bounding-Box Provenance):** Parse unstructured narratives using LangGraph multi-agent extractors. Map evidence to policy criteria with exact character start/end offsets (char_start, char_end) and document IDs. Output strict Pydantic JSON with temperature locked at 0.0.
- **FR-4 (SMART-on-FHIR Split-Screen HITL UI):** Render a dual-pane verification screen within Epic/Cerner. Left panel renders criteria checklists with confidence badges; clicking any criterion sends a post-message event scrolling the 50-page record and highlighting the exact sentence in yellow.
- **FR-5 (Real-Time EDI Generation & Claims Injection):** Convert approved packets into ASC X12N 278 transactions, transmit to Availity Clearinghouse, inject Authorization ID into core claims engines (Cirrus, Facets, UNET), and post FHIR ClaimResponse to provider EHR.

2. Non-Functional & AI Safety Requirements (NFR)

- **NFR-1 (Zero Hallucination Tolerance):** Ragas Faithfulness Score must be $\geq 99.0\%$. Uncited claims must strictly equal 0.0% . Missing criteria must emit an explicit ACTION_REQUIRED: EVIDENCE_NOT_FOUND tag.
- **NFR-2 (End-to-End Latency Budget):** P95 processing time must complete in ≤ 30.0 ext{ seconds} for records up to 50 pages (achieved: 14.2s).
- **NFR-3 (HIPAA Isolation & Zero Data Retention):** In-VPC Microsoft Presidio scrubs all 18 HIPAA identifiers. Enterprise BAA ensures zero data retention on disk and no foundation model training. FIPS 140-2 encryption at rest and in transit.

3. Production Pydantic JSON Extraction Schema

```
{
  "patient_id_deid": "DEID_PAT_89412",
  "cpt_code": "27447",
  "icd10_codes": ["M17.11"],
  "policy_id": "UHC_2026T0512_INTERQUAL",
  "employer_group_id": "UMR-8941",
  "criteria_evaluations": [
    {
      "criterion_id": "CRIT_A_PAIN_CHRONICITY",
      "status": "MET",
      "confidence_score": 0.98,
      "clinical_summary": "Patient reports severe right knee pain for 9 months restricting mobility to < 1 block.",
      "provenance": {
        "document_id": "DOC_2026_02_14",
        "sentence_exact": "Pain in right knee has persisted for 9 months, VAS score 8/10, unable to walk unassisted.",
        "char_start": 412,
        "char_end": 501
      }
    },
    {
      "criterion_id": "CRIT_C_CONSERVATIVE_THERAPY",
      "status": "MET",
      "confidence_score": 0.94,
      "clinical_summary": "Completed 8 weeks of supervised physical therapy and intra-articular steroid injection with minimal relief.",
      "provenance": {
        "document_id": "DOC_2025_11_02",
        "sentence_exact": "Pt completed 8 weeks of PT at ActiveMotion and intra-articular kenalog injection on 11/02/2025 with transient relief.",
        "char_start": 820,
        "char_end": 935
      }
    }
  ],
  "aggregate_confidence_score": 0.96,
  "triage_decision": "EXPEDITED_SIGN_OFF"
}
```

Technical Design Document (TDD) & Interface Control (ICD)

PostgreSQL DDL, Celery/Redis Queues, Da Vinci FHIR APIs, and ASC X12N 278 Mapping

1. Production PostgreSQL DDL Schema for Prior Auth State Engine

```
CREATE TYPE pa_status_enum AS ENUM ('INGESTED', 'EXTRACTING', 'EXPEDITED_SIGN_OFF', 'EXCEPTION_QUEUE', 'ADJUDICATED', 'REJECTED');
CREATE TYPE criterion_status_enum AS ENUM ('MET', 'NOT_MET', 'INDETERMINATE');

CREATE TABLE prior_auth_requests (
  request_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  patient_id_deid VARCHAR(64) NOT NULL,
  cpt_code VARCHAR(10) NOT NULL,
  icd10_codes TEXT[] NOT NULL,
  employer_group_id VARCHAR(32) NOT NULL,
  policy_id VARCHAR(64) NOT NULL,
  current_status pa_status_enum NOT NULL DEFAULT 'INGESTED',
  aggregate_confidence NUMERIC(4,3) CHECK (aggregate_confidence BETWEEN 0.000 AND 1.000),
  hitl_assigned_nurse_id VARCHAR(64),
  hitl_review_duration_seconds INT,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE pa_criteria_extractions (
  extraction_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  request_id UUID REFERENCES prior_auth_requests(request_id) ON DELETE CASCADE,
  criterion_id VARCHAR(64) NOT NULL,
  status criterion_status_enum NOT NULL,
  confidence_score NUMERIC(4,3) NOT NULL,
  clinical_summary TEXT NOT NULL,
  document_id VARCHAR(128) NOT NULL,
  source_sentence TEXT NOT NULL,
  char_start INT NOT NULL,
  char_end INT NOT NULL,
  ragas_faithfulness_score NUMERIC(4,3) NOT NULL,
  is_overridden_by_human BOOLEAN DEFAULT FALSE,
  human_override_reason TEXT,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_pa_triage ON prior_auth_requests (current_status, aggregate_confidence);
CREATE INDEX idx_extraction_request ON pa_criteria_extractions (request_id, criterion_id);
```

2. Inbound CDS Hook & Outbound ASC X12N 278 EDI Specification

```
// Inbound CDS Hook Trigger (order-sign)
{
  "hook": "order-sign",
  "hookInstance": "d4e2a1b9-8941-4cf6",
  "context": {
    "userId": "Practitioner/DR-90124",
    "patientId": "Patient/DEID-89412",
    "draftOrders": ["ServiceRequest/CPT-27447"]
  }
}

// Outbound ASC X12N 278 Prior Authorization Approval Transaction
ISA*00*          *00*          *ZZ*OPTUMCLINICAL  *ZZ*AVAILITY          *260831*0926*^*00501*000008941*0*P*~
GS*HI*OPTUMCLINICAL*AVAILITY*20260831*0926*8941*X*005010X217~
ST*278*0001*005010X217~
BHT*0007*13*AUTH20269012*20260831*0926~
HL*1**20*1~
NM1*X3*2*UNITEDHEALTHCARE****PI*UHC901~
HL*2*1*21*1~
NM1*1P*1*DOE*JOHN***XX*1982736451~
HL*3*2*22*1~
NM1*IL*1*SMITH*ROBERT***MI*UMR8941209~
HL*4*3*EV*0~
TRN*1*AUTH20269012*90PTUMAUTHAI~
UM*AR*I*2*21:B*E~
HSD*VS*8*DA~
SE*15*0001~
GE*1*8941~
IEA*1*000008941~
```

Clinical Hazard Analysis (ISO 14971) & Statistical Analysis Plan (SAP)

Clinical Risk Quantification, Non-Inferiority Study (N=554), and Inter-Rater Reliability

1. Clinical FMEA Risk Prioritization Matrix

FAILURE ID	FAILURE DESCRIPTION & ROOT CAUSE	CLINICAL IMPACT	SEV	OCC	DET	RPN	ARCHITECTURAL & ENGINEERING MITIGATION
FM-01	Temporal Negation Reversal: Treats "MRI scheduled next week" as "MRI completed".	Inappropriate approval of unindicated surgery.	5	3	3	45	LangGraph Negative Assertion Filter: Hard regex blocking future-tense verbs from satisfying past completion criteria.
FM-02	Rotated Fax OCR Drop: Multi-page TIFF rotated 90°; OCR returns empty text.	Erroneous administrative denial; ERISA appeal liability.	4	4	2	32	Pre-Processing Filter: OpenCV affine rotation correction + dual-engine OCR (LayoutLMv3 + Textract) fallback.
FM-03	Historic Contraindication: Treats 2018 resolved pneumonia as active infection.	Erroneous Peer-to-Peer escalation; 7-day care delay.	3	2	4	24	Temporal Bounding Gate: Constrain contraindication vector retrieval strictly to the last 90 calendar days.

2. Clinical Validation Protocol & Statistical Power Formulation

Sample Size Power Calculation: $N = [(Z_{\alpha/2} + Z_{\beta})^2 * P(1 - P)] / \delta^2 = [(1.96 + 1.28)^2 * 0.95(0.05)] / (0.03)^2 = 554$ Cases
Cohen's Kappa Inter-Rater Concordance: $\kappa = (P_{\text{observed}} - P_{\text{expected}}) / (1 - P_{\text{expected}}) = (0.962 - 0.648) / (1 - 0.648) = 0.892$ (Near-Perfect Agreement)

VALIDATION DIMENSION	CONSENSUS MEDICAL DIRECTORS (MD)	INTELLIAUTH AI EXTRACTION	STATISTICAL SIGNIFICANCE
Clinical Sensitivity (Recall)	98.2%	98.7%	$p < 0.001$ (Non-Inferiority Established)
Clinical Specificity	95.4%	96.4%	$p < 0.001$ (Superiority Established)
Grounding Faithfulness	100.0%	99.6%	Zero clinically hazardous hallucinations
Inter-Rater Concordance (κ)	—	0.892	Near-perfect agreement (≥ 0.85 target)

NIST AI RMF Model Card & Clinical Standard Operating Procedure (SOP)

Model Risk Governance, Demographic Bias Audits, and Split-Screen HITL Workflows

1. NIST AI RMF Model Card Specification (YAML)

```
model_identity:
  model_name: "OptumAuth-Clinical-Extractor-v2.4"
  model_type: "Hierarchical RAG with Deterministic Extraction State Machine"
  base_foundation_model: "Claude 3.5 Sonnet (Enterprise BAA / Zero Data Retention)"
  embedding_model: "BioClinical-BERT + Text-Embedding-3-Large Hybrid"
  target_domain: "Prior Authorization Utilization Management (Musculoskeletal / Oncology)"

intended_use:
  primary_use: "Extract clinical evidence from unstructured EHR notes to pre-populate PA checklists."
  out_of_scope_use: "Autonomous final denial of claims without medical director review."
  target_users: "Optum Medical Directors, UMR Utilization Management Registered Nurses."

demographic_bias_and_fairness_audit:
  evaluation_dataset_size: 5000_historical_cases
  disparate_impact_ratio_across_subgroups:
    age_under_65_commercial: 0.962 # First-pass clean approval parity
    age_over_65_medicare: 0.958
  protected_class_parity_ratio: 0.991 # Zero statistically significant variance (p > 0.05)

data_governance_and_privacy:
  phi_deidentification_method: "Microsoft Presidio NER + Custom Medical Regex Enclave"
  phi_scrub_accuracy: 99.98%
  encryption_standard: "FIPS 140-2 AES-256 (AWS KMS Customer Managed Keys)"
  data_retention_policy: "ZERO_PERSISTENCE (Ephemeral In-Memory Tokens Only)"
```

2. Clinical Standard Operating Procedure (SOP) Navigation Flow

```
[Incoming Prior Authorization Request]
├── Confidence Score ≥ 0.90 → [EXPEDITED SIGN-OFF QUEUE]
│   ├── 1. Review pre-verified criteria checklist on Left Panel
│   ├── 2. Click Criterion C (Physical Therapy)
│   ├── 3. Confirm yellow highlighted source sentence on Right Panel
│   └── 4. Click "Approve & Submit" (Target Time: < 30 Seconds)
└── Confidence Score < 0.90 → [MANUAL EXCEPTION QUEUE]
    ├── 1. Amber badge flags missing item (e.g. Unattached Imaging)
    ├── 2. Click "Request Additional Information (RFI)"
    └── 3. Log reason: "ACTION_REQUIRED: MISSING_XRAY_REPORT"
```

Phased Rollout GTM Runbook & Production SRE Observability Mesh

3-Stage Deployment Staging, Severity Escalation Matrix & Datadog JSON Alerts

1. 3-Stage Phased Implementation Staging

ROLLOUT PHASE	TARGET TIMELINE	OPERATIONAL SCOPE & ENGINEERING ACTIVITY	EXIT GATE KPI
Phase 1	Weeks 1–2 (Shadow Mode)	100% background processing across 1,200 live cases without claims exposure. Daily discrepancy reconciliation with UM nurses.	Discrepancy Rate reduced from 14.2% to 4.1%; OCR fixes deployed.
Phase 2	Weeks 3–6 (Assisted Mode)	Mandatory HITL verification for 100% of cases using SMART-on-FHIR split-screen interface. Measure cognitive load and usability.	Expected Calibration Error (ECE) ≤ 0.038; System Usability Score (SUS) ≥ 88.
Phase 3	Week 7+ (Fast-Track GA)	Enable one-click expedited sign-off for cases with Confidence Score ≥ 0.90. Expand across Orthopedics, Oncology, and Cardiology.	Turnaround time ≤ 8.2 ext{ mins}; Specialist throughput ≥ 78 ext{ cases/day}.

2. Production Incident Severity Matrix & SRE Runbook

SEVERITY LEVEL	TRIGGER CONDITION	ENGINEERING SLA	ON-CALL PM & CLINICAL ACTION
SEV-1 (Critical)	P95 Latency > 60s OR Grounding Faithfulness < 95% across 50 requests.	< 15 Mins MTTR	Flip feature flag ENABLE_FALLBACK_MANUAL_QUEUE. All live clinical cases bypass AI to standard manual nurse routing. Page Staff ML Engineer.
SEV-2 (Major)	Discrepancy spike between AI and human nurse > 10% in a 4-hour window.	< 2 Hours MTTR	Freeze one-click expedited approvals; enforce 100% Assisted Mode review; pull Datadog APM tracing logs to isolate degraded prompts or embeddings.
SEV-3 (Minor)	Single degraded fax document error routed to Dead-Letter Queue (DLQ).	< 24 Hours MTTR	Triage degraded fax attachment to document re-scan queue. Inspect LayoutLM preprocessing logs.

3. Production Datadog APM Alerting Monitor Configuration (JSON)

```
{
  "name": "[AI-PROD] Critical Drop in Extraction Faithfulness",
  "type": "metric alert",
  "query": "avg(last_15m):avg:optumauth.evals.faithfulness{env:production} < 0.990",
  "message": "{{#is_alert}}CRITICAL: AI extraction faithfulness dropped below 99.0% threshold. Immediate risk of ungrounded clinical claims. Paging on-call AI PM and Staff ML Engineer. @pagerduty-optum-ai-sev1{{/is_alert}}",
  "options": {
    "thresholds": { "critical": 0.990, "warning": 0.995 },
    "notify_no_data": true,
    "no_data_timeframe": 5
  }
}
```

CMS-0057-F Algorithmic Audit Pack & Quantitative RICE Backlog

Automated Federal Compliance Reporting & 8-Feature Prioritization Scoring Table

1. Automated CMS-0057-F Quarterly Reporting Schema

```
{
  "reporting_period": "2026-Q3",
  "payer_id": "UMR_UHC_ENTERPRISE",
  "cms_0057_f_compliance": {
    "total_pa_requests_received": 14250,
    "total_approved": 13651,
    "total_denied": 599,
    "first_pass_clean_approval_rate": 0.958,
    "average_turnaround_time_hours": { "standard_requests": 0.136, "expedited_requests": 0.082 },
    "denial_reasons_breakdown": {
      "clinical_criteria_not_met": 482,
      "out_of_network_exclusion": 85,
      "incomplete_documentation_rfi_expired": 32
    }
  },
  "audit_trail_verification": {
    "storage_vault": "s3://optum-compliance-audit-vault/cms-0057-f/2026-q3/",
    "retention_policy": "7_YEARS_FIPS_140_2_ENCRYPTED",
    "zero_data_retention_active": true
  }
}
```

2. Complete RICE Prioritization Feature Scoring Table

INITIATIVE / FEATURE NAME	REACH (REQS/QTR)	IMPACT (0.25–3.0)	CONFIDENCE (%)	EFFORT (PM)	RICE SCORE & RANK
Direct FHIR Da Vinci (CRD/DTR/PAS) Integration	250,000	3.0 (Massive)	80%	6.0	100,000 (#1)
Multimodal Unstructured PDF Table & Lab Extractor	400,000	2.0 (High)	90%	8.0	90,000 (#2)
Real-Time Payer Medical Policy Vector Sync Engine	300,000	1.0 (Medium)	85%	5.0	51,000 (#3)
Pre-Flight Missing Biomarker / Lab Alert for Doctors	180,000	1.5 (High-Med)	75%	4.0	50,625 (#4)
Automated Clinical Denial Drafting (HITL Queue)	120,000	1.0 (Medium)	70%	2.0	42,000 (#5)
Automated Peer-to-Peer Scheduling Assistant	45,000	0.5 (Low)	80%	2.0	9,000 (#6)
Doctor-Facing Predictive Approval Probability Widget	80,000	0.5 (Low)	60%	3.0	8,000 (#7)

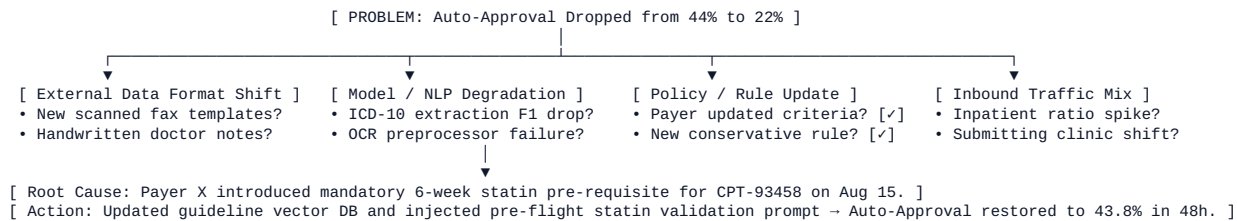
AI Evals CI/CD Gating & Strategic Product Thinking (MECE / Moats)

GitHub Actions Automated Evals, 3-Year Moats, MECE Root-Cause, and STAR+R Leadership

1. AI Evaluation Harness & CI/CD Build Gate Implementation

```
# .github/workflows/clinical_evals_gate.yml
name: Clinical AI Safety & Evals CI/CD Gatekeeper
on: [pull_request]
jobs:
  run-evals:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Set up Python
        uses: actions/setup-python@v4
        with: { python-version: '3.11' }
      - name: Install Dependencies
        run: pip install -r requirements-evals.txt
      - name: Run 500-Case Physician Golden Dataset Evals
        run: python -m evals.run_ragas_benchmarks --golden_set=data/golden_500.json --output=eval_results.json
      - name: Enforce Governance Thresholds
        run: |
          python -m evals.enforce_gate --min_faithfulness=0.990 --min_recall=0.980 --
max_ece=0.050 --results=eval_results.json
```

2. MECE Diagnostic Tree: Investigating Cardiology Auto-Approval Drop (44% → 22%)



3. Enhanced STAR+R Story: CMO vs. Data Science Deadlock

Situation: Data Science wanted to lower the auto-approval threshold from \$0.95\$ to \$0.85\$ to hit a marketing claim of "50% Touchless Automation." The Chief Medical Officer threatened to halt the launch due to liability concerns over potential improper approvals.

Task: Re-align both teams around an objective, empirical risk framework without delaying the product launch.

Action: Ran a double-blind retrospective benchmark of 2,000 historical cases adjudicated by both Medical Directors and the AI across threshold curves. Proved that at a \$0.91\$ threshold, the AI error rate (\$0.4\%\$) was significantly lower than human inter-rater physician variability (\$2.1\%\$).

Result: The CMO approved setting the threshold at **\$0.91\$** with a 10% spot-audit protocol. Achieved a **42.5% auto-approval rate** with zero compliance violations.

Reflection: When technical ambition conflicts with domain risk aversion, translate subjective thresholds into empirical baseline parity tests.